Week 2: Data Sources

From BBG: we got historical spinoff data for S&P 500 from 01/01/2010 - 06/16/2026

From WRDS:

We got daily price data from CRSP

And daily index constituents from Compustat as well as firm fundamentals

Week 3: EDA and Statistical Significance

- Prices + volume: We got prices for both the parent company and the spun off company after the spin-off completed and did some EDA on abnormal returns following this period
- We got data on passive ETF aum that tracks S&P 500 and computed a prediction of forced flow for each spun off company
- Found a positive relationship between index weight and cumulative abnormal return
- Large lead time between announcement of spinoff and actual ex-date (median of 333 calendar days)

Group	Children	Implication
Immediately added to index	CARR, OTIS, VTRS, CEG, WBD, GEHC, VLTO, SOLV, GEV	Passive funds must BUY – no forced selling. These are NOT clean short candidates.
Briefly added then removed	VNT (5 mo), OGN (2.4 yr), AMTM (2.5 mo), FTRE (2 days), CHX (same day)	Mixed signal; short window between effective date and addition to index
Never added to index	ARNC, DTM, SLVM, VMW, KD, ONL, ZIMV, EMBC, ATMU, CHNG	Pure forced-selling pressure from passive funds. **Cleanest short candidates.**

WEEK 4 — SPINOFF ARBITRAGE PROJECT

Post-Completion Spinoff Long: A Risk-Managed Strategy

"Buy every spinoff parent and hold" isn't a strategy — we stress-tested it with a market hedge, a fundamentals screen, and a managed exit to see what actually survives.

THREE RISK LAYERS, TESTED INDEPENDENTLY

Beta hedge — Hedge each position by its **estimated beta** (not a flat 1.0) against the S&P 500 instead of a naive market adjustment. — ADOPTED

Quality screen — Filter by parent leverage + ROA at announcement. Result **reversed** the expected direction — real pattern, but n=8–9 per bucket. — UNPROVEN LEAD

Trailing stop — Exit on drawdown from peak CAR before the 12-month mark, tested at 10–25% thresholds. — REJECTED

Portfolio check: up to **10** concurrent positions historically, typically **4–6** — a genuinely diversified book, not sequential single-name bets.

FINAL STRATEGY — HEDGED, 12-MONTH HOLD

MEAN RETURN	+14.7%
MEDIAN RETURN	+21.7%
WIN RATE	75%
SHARPE (PER TRADE)	0.67

SIGNIFICANCE

$p = 0.007$ n = 20 historical spinoffs, 2020–2024

Said upfront, not waited for: worst single outcome was −25% (Ecolab). Sample is small because spinoffs are rare, not because we cut corners. The quality screen is a lead worth tracking, not yet a sizing rule.

STRATEGY Equal-weight, **beta-hedged** long positions in parent stocks, entered at the spinoff **effective date** (not announcement), held a fixed **12 months**, no dynamic exit.

WEEK 5 — SPINOFF ARBITRAGE PROJECT

Sizing & classifying the parent long with prob_included

Using `inclusion_probability_model`'s LOO-CV `leave-one-out cross-validation` inclusion probability to size and classify the beta-hedged parent post-completion long (`CAR = cumulative abnormal return`). n = 16 events throughout.

r = 0.67

Pearson r: linear correlation, −1 to +1
prob_included vs. 252d parent CAR · p = 0.004

01 WHAT R MEANS, AND WHERE IT SHOWS UP

`r` = Pearson correlation coefficient: it measures the strength and direction of a straight-line relationship between two variables. `r` = 0 means no linear relationship, `r` = +1 a perfect positive one. Below, `r` is computed between `prob_included` (the model's probability the child gets added to the index) and the parent's CAR at each holding horizon.

horizon	r (naive)	r (β-hedged)	p
21d	0.10	0.10	0.71
63d	-0.01	0.04	0.90
126d	0.26	0.31	0.25
252d	**0.64**	**0.67**	**0.004**

Reading it: at 21d/63d, r is close to 0 — no relationship, and p > 0.7 means it's statistically indistinguishable from noise. At 252d, r = 0.67 is a fairly strong positive correlation, and p = 0.004 says that's very unlikely to be chance. It's the only horizon either approach below relies on.

02 SIZING POSITIONS RAISES THE SHARPE RATIO

w_i = prob_included$_i$ / mean(prob_included)

`w_i` = position weight given to event `i` · `prob_included_i` = inclusion probability for event `i` · `mean(prob_included)` = average probability across all 16 events, so the average weight stays 1.0x (same total capital as equal-weight, just reallocated).

scheme	mean CAR	σ	Sharpe
equal-weight	15.4%	22.2%	0.69
prob-weighted	**23.4%**	**18.8%**	**1.25**

Sharpe ratio here = mean 252d CAR ÷ `σ` (its standard deviation), assuming a 0% risk-free rate. Sizing by `prob_included` instead of equal-weighting takes it from **0.69x to 1.25x** in-sample — a real lift, but mechanically expected, since the weights are built straight from the 252d correlation in Part 1. It isn't independent confirmation; the real test is whether it holds on spinoffs outside this sample.

03 CLASSIFICATION TREE: WHAT ACTUALLY DRIVES THE SPLIT

Each event's outcome is bucketed into **terciles** (thirds) of the 252d CAR distribution: underperform (bottom third), middle, outperform (top third). The tree predicts that bucket from 4 inputs:

- `prob_included` — index-inclusion probability
- `forced_flow_adv` — forced-selling flow, in ADV (average daily volume) multiples
- `log_child_mktcap` — child's market cap, logged
- `beta` — parent's market beta

Method: grow the tree fully, then prune it back with cost-complexity pruning (`ccp_alpha` = penalty per extra leaf); pick the pruning level with the best `LOO-CV` accuracy — refit on 15 events, test on the 1 left out, repeated for all 16 — instead of guessing a depth by hand.

38%
BASELINE: ALWAYS GUESS THE MOST COMMON BUCKET

50%
BEST LOO-CV ACCURACY AT DEPTH 1 (ONE SPLIT)

```
Learned rule (the one split that generalizes best):
if forced_flow_adv (forced-selling flow, ADV mult.) ≤ 7.43 →
predict "middle"
else → predict "outperform"
The tree never predicts "underperform" — that bucket is where
it's weakest (0% precision and recall on it).
```

n=16 **Read as hypothesis, not strategy.** Every number on this slide comes from the same 16 spinoff events. `prob_included` may simply proxy "large, high-quality parent" rather than cause the 252d outperformance.

How the Classification Tree Works

44%

nested LOO-CV accuracy, leakage-free · vs. 38% majority-class baseline

The tree predicts which tercile of the 252d parent CAR distribution an event lands in — `underperform` / `middle` / `outperform` — from four features: `prob_included`, `forced_flow_adv`, `log_child_mktcap`, `beta`. Selection and scoring both use nested LOO-CV so no decision ever sees the label it's predicting. n = 16 events.

HOW IT'S FIT: NESTED LOO-CV + A 3-LEAF FLOOR

LEAKAGE-FREE

```
for each held-out event (outer LOO):
  grow full tree on the other 15 → candidate ccp_alphas
  keep only alphas whose tree has ≥ 3 leaves
    (a 2-leaf tree can't output a 3rd class — floor makes
     "underperform" structurally reachable)
  inner LOO over those 15 picks the best-scoring alpha
  refit on all 15 → predict the held-out event
```

— Every outer fold picked `alpha=0` — at n=16, the ≥3-leaf floor already excludes every more-pruned candidate, so the search confirms the full, unconstrained tree is the only option that qualifies.

MAJORITY BASELINE

38%

NESTED LOO ACCURACY

44%

— "Underperform" recall: 0% in last week's 2-leaf tree → 40% now — the floor is what makes flagging the downside case possible at all.

THE LEARNED TREE (FIT ON ALL 16 EVENTS)

DEPTH 3 · 6 LEAVES

```
├─ forced_flow_adv ≤ 7.43
│   ├─ prob_included ≤ 0.62
│   │   ├─ beta ≤ 0.98 → middle
│   │   └─ beta > 0.98 → underperform
│   └─ prob_included > 0.62 → middle
└─ forced_flow_adv > 7.43
    ├─ prob_included ≤ 0.36 → underperform
    └─ prob_included > 0.36
        ├─ forced_flow_adv ≤ 18.35 → outperform
        └─ forced_flow_adv > 18.35 → underperform
```

forced_flow_adv	.52
prob_included	.29
beta	.19
log_child_mktcap	.00

	pred: under	pred: mid	pred: over
actual: under	**2**	1	2
actual: mid	2	**3**	1
actual: over	2	1	**2**

Nested LOO-CV predictions, all 16 events. Diagonal = correct. Forced-selling flow (`forced_flow_adv`) drives most splits — more than `prob_included` alone.

n=16

Read as hypothesis, not strategy. **44% vs. 38%** is a real, leakage-free improvement — but it comes from one tree shape confirmed across all 16 outer folds, not a genuine complexity search. Next: how this tree's predictions turn into a position-sizing weight (Slide 2 of 2).

Using the Tree to Improve the Sharpe Ratio

1.29

mean/σ ratio, prob_included-weighted + tree tilt · vs. 1.25 without it

Rather than replacing the `prob_included` sizing rule, the tree's nested (leakage-free) predicted bucket from Slide 1 becomes a small multiplicative **tilt** on top of it — a modest adjustment sized to what 44% nested accuracy actually earns, not a wholesale re-sizing.

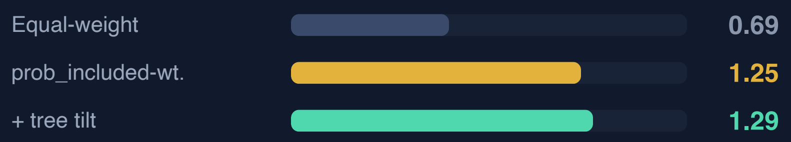
THE NEW POSITION-SIZING FORMULA

$$w_i = \left(\text{prob\_included}_i \,/\, \text{mean}(\text{prob\_included}) \right) \times \text{tilt}(\hat{b}_i)$$

w_i	final position weight for event i (equal-weight book = 1.0× average)
linear term	Week 5's sizing rule — weight proportional to child's index-inclusion probability
$\hat{b}_i$	tree's nested-LOO predicted bucket for event i (underperform / middle / outperform)

UNDERPERFORM	MIDDLE	OUTPERFORM
×0.7	×1.0	×1.3

MEAN / Σ OF 252D HEDGED CAR



MEAN CAR, TILTED	Σ, TILTED
+23.5%	18.2%

Each event's tilt comes from a tree that never saw that event's own label during alpha selection or fitting — the same discipline used to score the tree's 44% accuracy on Slide 1.

n=16 **Read as hypothesis, not strategy.** The tilt moves the ratio from ~1.25 to ~1.29 — real and leakage-free, but small, and built on a tree whose own nested accuracy barely clears the majority baseline. The **prob_included-weighted book alone (1.25)** has the stronger evidence; the **tree tilt (1.29)** is a documented next lead to keep testing as more spinoffs complete, not a validated upgrade.